12TH BIOLOGY · CHAPTER 24 · FULL CHAPTER NOTES

Evolution

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What is the theory of creationism?

According to creationism (the “theory of intelligent design”), all organisms were created as they exist today. Each type of organism was created individually and did not go through any change throughout history.

Q.2 [Short] What was the idea of spontaneous generation?

Early people believed that living creatures can develop from non-living matter. This was the idea of spontaneous generation. Later experiments disproved it, but such experiments could not negate creationism.

24.1 THE CONCEPT OF EVOLUTION

Q.3 [Short] Define evolution. What were the basic speculations of its early concept?

Evolution is the gradual change in the genetic makeup (allele frequencies) of a population over successive generations, which may lead to the formation of new species. In its earliest form, the concept had these basic speculations:

- The universe and organisms did not always exist in their present form.
- Today’s organisms were not created at once.
- They are the result of innumerable changes from the lower to the higher form.

Q.4 [Short] How does evolution take place due to variations and natural selection?

Evolution takes place due to **variations** in organisms and **natural selection**. Organisms with beneficial traits are selected to produce the next generations, while individuals with disadvantageous traits cannot produce next generations. As a result, the percentage of individuals with better traits increases in the population.

24.2 THEORY OF EVOLUTION – DARWINISM

Q.5 [Short] Who was Charles Darwin? What was the voyage of HMS Beagle?

Charles Darwin (1809–1882) was an English scientist and the founder of the modern theory of evolution. In 1831 he travelled as a naturalist on a scientific expedition on His Majesty’s Ship, HMS Beagle, a 5-year voyage around the world. In 1859 he published “On the Origin of Species”.

Q.6 [Long] Describe the observations made by Darwin during his voyage.

- **Diversity and fossils:** In tropical rainforests and other new habitats he saw a great diversity of plants and animals. He found rocks containing fossils, which suggested that continents and oceans had changed dramatically over time.
- **Extinct mammals:** He dug up fossils of gigantic extinct mammals; organisms looked very different in the past. This suggested that living things change over time.
- **Galápagos Islands (16 small islands):** Plants and animals on different islands were very different, e.g., giant tortoises had saddle-shaped shells on one island and dome-shaped shells on another. Many of them resembled the plants and animals of South America.
- **Finches:** Different species of finches had different beaks. The sizes and shapes of the beaks were adaptations for feeding on completely different kinds of food on different islands.

His explanation: The origin of new species and adaptation are closely related processes. A new species arises from an ancestral form by the gradual accumulation of adaptations to a different environment, separated from the original habitat. Over many generations the two populations become dissimilar enough to be separate species.

Q.7 [Short] What did Darwin observe in the finches of the Galápagos Islands?

Different species of finches had different beaks. The differences in size and shape of the beaks were adaptations for feeding on completely different kinds of food on different islands. From this he concluded that the origin of new species and adaptation are closely related processes.

Q.8 [Short] What was the contribution of Alfred R. Wallace?

Alfred R. Wallace (1823–1913) was a British naturalist and biologist who developed his own theory on the mechanism of evolution, very similar to Darwin's. He wrote a letter about it to Darwin. When both theories were presented to the Linnaean Society in 1858, Wallace agreed to attribute the theory to Darwin and convinced him to publish his book.

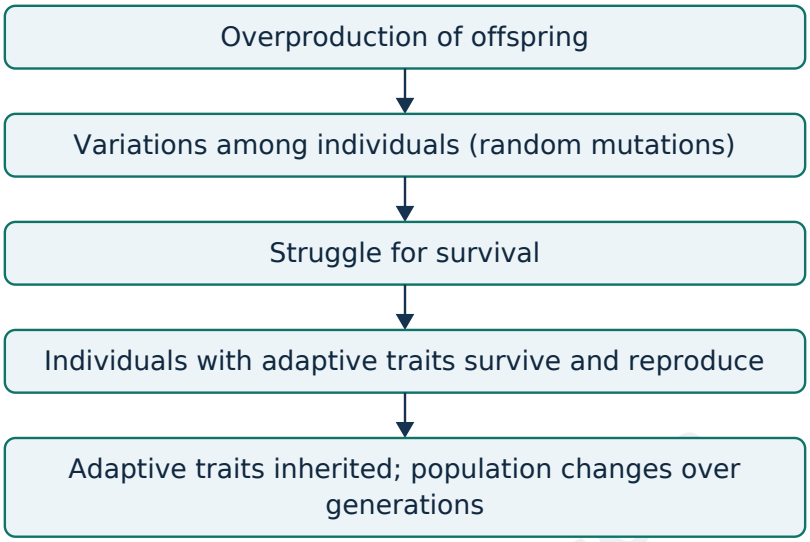
The textbook prints the year of Wallace's letter as "1957"; this is a misprint, since Wallace died in 1913 and the theories were presented in 1858.

Q.9 [Long] Explain the mechanism of evolution by natural selection as proposed by Darwin.

1. **Overproduction:** All organisms have a far greater reproductive potential than is ever realized. For example, a female sea star releases about 1 million eggs each season; if all developed into adults, sea stars would fill the oceans in a few generations.
2. **Variations:** Genetic variations arise by random mutations. Seldom are any two individuals exactly alike. Variations may be advantageous, harmful or neutral. Inherited variations can be passed on to offspring.
3. **Struggle for survival:** Many more offspring are produced than resources can support, so individuals struggle for survival. Individuals with disadvantageous variations die or cannot reproduce; those with advantageous variations reproduce

successfully. Such variations are called **adaptive traits**.

4. **Inheritance of adaptive traits:** Adaptive traits are inherited by the next generations. Disadvantageous traits become less frequent in the population and are eventually eliminated.



Natural selection as proposed by Darwin

Q.10 [Short] What is meant by overproduction? Give an example.

All organisms have a far greater reproductive potential than is ever realized. For example, a female sea star releases about 1 million eggs each season. If all were fertilized and grew into adults, a few generations would fill up the oceans.

Q.11 [Short] What are adaptive traits?

In the struggle for survival, individuals with advantageous variations remain successful in reproduction. Such advantageous variations are called adaptive traits. They are inherited by the next generations.

Q.12 [Short] What is Neo-Darwinism?

In the early 20th century, Darwinism was integrated with modern genetics to develop Neo-Darwinism. It defines evolution more precisely as “a change in the frequency of alleles in a population or in the total genetic makeup of a population (the gene pool)”. It is based on the Hardy-Weinberg theorem.

Q.13 [Short] Define gene pool, allele frequency and genotype frequency.

Term	Meaning	Example
Gene pool	Total number of all genes for a trait in a population	1000 genes for flower colour in a population
Allele (gene) frequency	Number or proportion of a specific allele in the population	600 of 1000 genes are for red: frequency 0.6 (60%)
Genotype frequency	Number or proportion of individuals with a particular genotype	40 of 100 plants are RR: frequency 0.4 (40%)

Q.14 [Long] State the Hardy-Weinberg theorem and its conditions.

The **Hardy-Weinberg theorem** was presented in 1908 by the English mathematician Godfrey Hardy and the German physician Wilhelm Weinberg. It states that the frequencies of alleles and genotypes in a large, randomly mating population remain constant from generation to generation, if no evolutionary forces are acting.

If any of the following conditions is not followed, evolution takes place:

1. The population size must be large.
2. Mating within the population must be random; every individual must have an equal chance of mating with any other individual.
3. Individuals must not migrate into or out of the population.
4. Mutations must not occur; if they do, mutations from the wild-type allele to the mutant allele must equal mutations back to the wild type.
5. All genotypes must have an equal chance of survival and reproduction.

Q.15 [Long] Describe the factors that can change allelic frequencies.

Genetic drift (neutral selection): When chance events influence the frequencies of genes in a population, the change is called genetic drift. Unlike natural selection, these changes occur by chance, not fitness, and are most obvious in small populations. Example: a fire reduces a rabbit population from 100 to 10; the survivors happen to be mostly brown, not because they were better adapted but by chance. The resulting population has lower genetic diversity.

Gene flow: Changes in gene frequency due to the migration of individuals are called gene flow. The entry of individuals from another population (immigration) can introduce new mutations and instantly change gene frequencies; individuals leaving (emigration) also change them.

Mutation: Mutations are the origin of all new genes and a source of variations that may prove adaptive. They ensure that variations are present which help a group survive environmental shocks. Most mutations are deleterious; some may be neutral or harmful in one environment and helpful in another. **Mutational equilibrium** (mutation from wild type to mutant balanced by mutation back) rarely exists. **Mutation pressure** is a measure of the tendency for gene frequencies to change through mutation.

Q.16 [Short] Why is genetic drift called neutral selection?

Because the change in gene frequencies is due to chance events and not to fitness. The survivors are not better adapted; they are simply lucky.

Q.17 [Short] What are mutational equilibrium and mutation pressure?

Mutational equilibrium exists when mutation from the wild-type allele to a mutant form is balanced by mutation from the mutant back to the wild type; the effect on allelic frequency is the same as if no mutation occurred. It rarely exists. **Mutation pressure** is a measure of the tendency for gene frequencies to change through mutation.

Q.18 [Short] Why are mutations important for evolution?

Mutations are the origin of all new genes and a source of variations that may prove adaptive. They make sure that variations are present which allow a group to survive environmental shocks more successfully.

24.3 EVIDENCES OF EVOLUTION

Q.19 [Long] How does biogeography provide evidence of evolution?

Biogeography is the study of the geographic distribution of plants and animals. It shows that life-forms in different parts of the world have distinctive evolutionary histories.

- **Similar groups separated by vast barriers:** Obvious similarities suggest a common ancestry, while obvious differences result from millions of years of independent evolution. Example: the African lion is very similar to the mountain lion of America.
- **Different organisms in similar environments:** Organisms separated by geographical barriers are often very different in spite of similar environments. Example: many animals of Australia and Tasmania are very different from animals in any other part of the world; many species of kangaroos are the major herbivores there.
- **Islands:** Islands often have relatively few but unique species, due to island colonization and subsequent evolutionary events.

Q.20 [Long] Describe the evidence of evolution from comparative anatomy.

Divergent evolution: Body parts that are similar in structure but different in function are called **homologous structures**, and the phenomenon is **homology**. Homologous structures reveal divergent evolution: the organisms have a common ancestor, evolved into different types and used their structures for different functions. Example: vertebrate forelimbs have a common arrangement of similar bones, although their functions vary: flight (birds, bats), swimming (whales, dolphins), running (horses), climbing (arboreal lizards) or swinging from tree branches.

Convergent evolution: Organs that are similar in function but differ in structure are called **analogous organs**. They indicate convergent evolution, in which different types of organisms have evolved from different ancestors in a common habitat. Example: the wing of a bird and the wing of an insect are both adaptations for flight, but they are not homologous. The similarity only reflects that, to fly, an animal needs a broad, flat gliding surface.

Vestigial organs: Organisms often retain structures that have lost their usefulness. These poorly developed structures are called vestigial structures. Example: boa constrictors and pythons have minute remnants of hindlimb (pelvic) bones left over from their reptilian ancestors. Such remnants of once-useful structures are clear indications of change, i.e., evolution.

Q.21 [Short] Differentiate between divergent and convergent evolution.

Divergent evolution	Convergent evolution
Shown by homologous structures.	Shown by analogous structures.
Structures are similar in structure but different in function.	Structures are similar in function but different in structure.
Organisms have a common ancestor (structure inherited from it).	Organisms have evolved from different ancestors (structure not inherited from a common ancestor).
Organisms evolve into different types and use the structures for different functions.	Different organisms in a common habitat develop structures for the same function.
Example: forelimbs of birds, bats, whales, horses.	Example: wing of a bird and wing of an insect.

Q.22 [Short] What are vestigial structures? Give examples from the human body.

Vestigial structures are poorly developed structures that have lost their usefulness. About 90 are present in the human body, e.g.:

- Vermiform appendix: a vestige of the cecum.
- Coccyx or tailbone: the remnant of the tail.
- Wisdom teeth: vestigial third molars that human ancestors used for grinding plant tissue.
- Ear muscles: minimally developed and non-functional, though some people can move their ears.

Q.23 [Long] Describe the evidence of evolution from molecular biology.

Just as animals have homologous structures, they also have **homologous biochemicals**.

- Structure and function are ultimately based on the genetic blueprint (DNA) found in all living animals.
- Related animals have similar DNA, derived from their common ancestor.
- DNA carries the codes for proteins, so related animals also have similar proteins.
- Scientists extract and analyse the DNA and proteins of different organisms. By looking for dissimilarities in related proteins and DNA, and by assuming relatively constant mutation rates, they estimate the time elapsed since evolution from a common ancestral molecule.

Q.24 [Short] How have microorganisms evolved resistance to antibiotics?

Antibiotic resistance means that bacteria are able to survive exposure to an antibiotic. It may be due to mutations in bacterial genes. A gene for antibiotic resistance, which evolved via natural selection, passes to the next generations of bacteria.

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24.4 EVOLUTION FROM PROKARYOTES TO EUKARYOTES

Q.25 [Short] When did prokaryotes and eukaryotes originate?

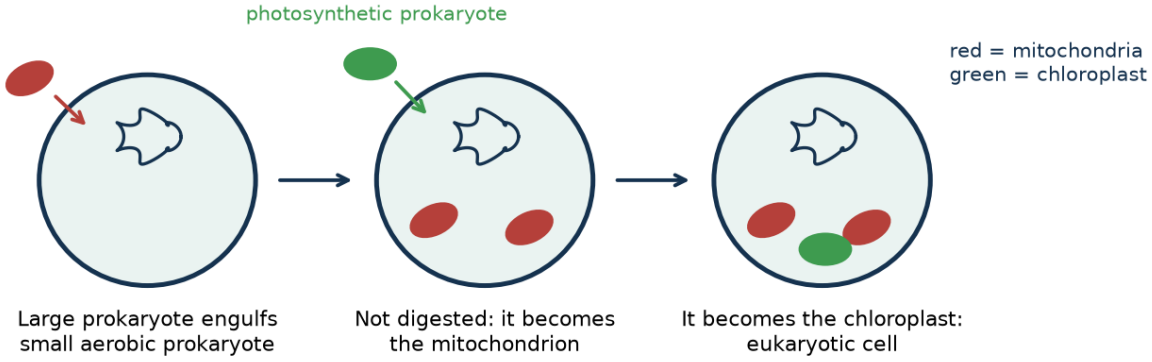
The earliest organisms were prokaryotes, which originated in a very unstable environment from the reactions of organic molecules about 3.5 billion years ago. About 2 billion years ago, eukaryotes evolved from prokaryotes.

Q.26 [Long] Describe the endosymbiotic theory.

The **endosymbiotic theory**, proposed by **Lynn Margulis**, is the most acceptable theory for the evolution of eukaryotes from prokaryotes (about 2 billion years ago).

1. Large prokaryotes engulfed small prokaryotes.
2. The small prokaryotes were not digested. They lived within the large cells and evolved into organelles. Both cells benefited (symbiotic relationship).
3. Some small prokaryotes could carry out aerobic respiration. They supplied energy to themselves and to the large cell, and became the **mitochondria**.
4. Other small cells could use sunlight to make food, which they shared with the large cell. They became the **chloroplasts**.
5. Flagella may have been derived through the ingestion of prokaryotes similar to spirochetes.

The eukaryotic cells so formed were powerful and efficient. This enabled them to evolve new characteristics such as multicellularity, cell specialization and large size, which led to the diversity of animals, plants and fungi.



Endosymbiotic origin of mitochondria and chloroplasts

Q.27 [Short] Give the evidence for the endosymbiotic theory.

- Mitochondria (and chloroplasts) are of the same size as prokaryotic cells.
- They have their own DNA, organized in a circular chromosome like a prokaryotic chromosome.
- They reproduce by binary fission like prokaryotes; eukaryotic cells reproduce by mitosis.

- Their membrane composition is more similar to prokaryotic membranes than to eukaryotic membranes.

Q.28 [Short] What is the membrane invagination theory?

According to this theory, the cell membrane of a prokaryotic cell invaginated into the cytoplasm. It gave rise to membranous structures, including the nucleus and endoplasmic reticulum, inside the prokaryotic cell, which resulted in the formation of a eukaryotic cell.

The last sentence in the textbook (p.222) reads "formation of a prokaryotic cell"; the sense requires "eukaryotic cell".

24.5 LAMARCKISM

Q.29 [Long] Describe the theory of evolution proposed by Lamarck.

The French naturalist **Jean-Baptiste Lamarck** (1744–1829) published the book *Philosophie Zoologique* (Zoological Philosophy) in 1809. In it he presented the idea of progression (change) known as **Lamarckism**: organisms changed from the simplest to complex, and changes occur in organisms when they adapt to the environment.

1. **Use and disuse of organs:** When an organism faces changes in its environment, it begins to use some organs more and disuses others. Organs used more frequently develop and become strong; organs not properly used deteriorate or become weak. Such changes, due to long-term use or disuse, are **acquired characters**.
2. **Inheritance of acquired characters:** The characters acquired during the lifetime of an organism are passed on to its offspring. Inherited from generation to generation, they ultimately modify the organism to cope better with the changes in the environment.

Q.30 [Short] What are acquired characters?

The changes in body parts that occur due to long-term use or disuse are called acquired characters. Organs used more frequently become strong; organs not used become weak.

Q.31 [Long] Explain the evolution of the giraffe neck according to Lamarckism.

1. The ancestors of the giraffe did not have long necks. They ate grass or the leaves of small plants.
2. The environment changed, and small plants were no longer available.
3. Some individuals began stretching their necks to reach the leaves of tall trees.
4. By constant stretching, their necks became strong and long during their lifetime: an acquired character.
5. When they reproduced, this acquired character was inherited; the young were born with slightly longer necks.
6. These individuals continued the same practice, and their necks became even stronger and longer.

7. After the character was inherited over many generations, the generation with long necks evolved.

Q.32 [Short] State the drawbacks of Lamarckism.

After the mechanisms of inheritance were discovered, scientists disproved Lamarckism on the following bases:

1. Acquired characters have no genetic basis and therefore cannot be inherited by the next generations.
2. Organs are not modified by the wish or requirements of an organism.

Q.33 [Short] Was Lamarck criticized for the idea of evolution or for his mechanism?

When Lamarckism was proposed, no evidence was available that could explain the mechanism of evolution. So Lamarck was criticized more for advocating ideas of evolutionary change than for the mechanism he proposed. After the inheritance mechanisms were discovered, the mechanism itself was disproved.

Q.34 [Short] Differentiate between Lamarckism and Darwinism.

Lamarckism	Darwinism
Changes arise by use and disuse of organs.	Variations arise by random mutations.
Acquired characters are inherited.	Inherited variations are selected by nature.
Not accepted; acquired characters have no genetic basis.	The most accepted theory of evolution.

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